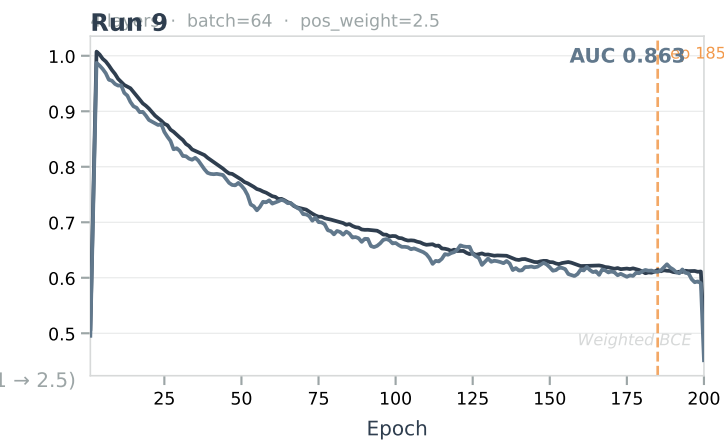
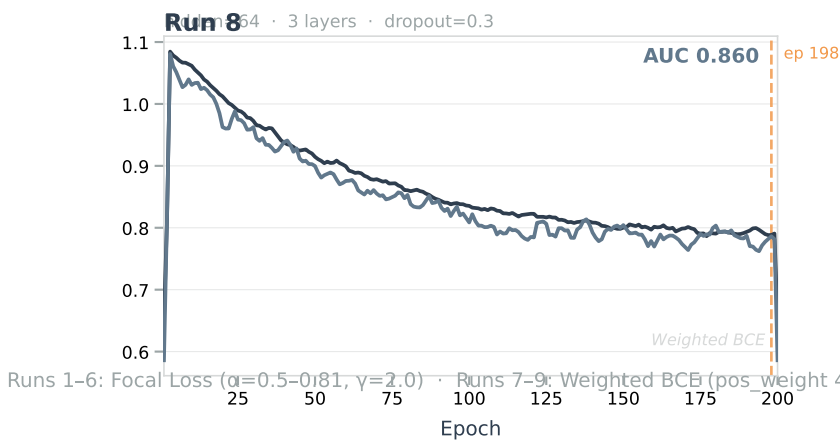
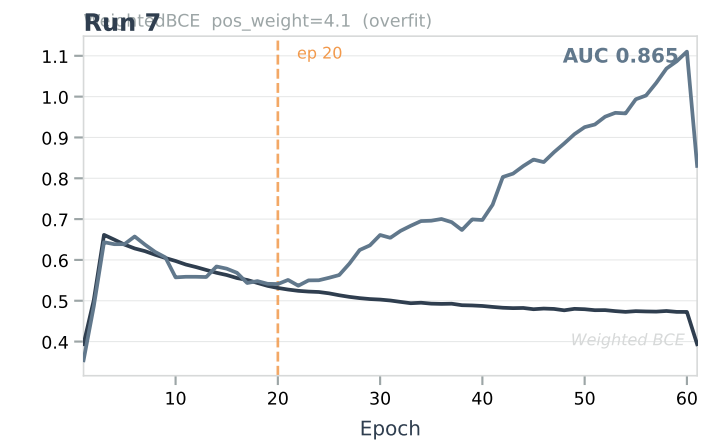
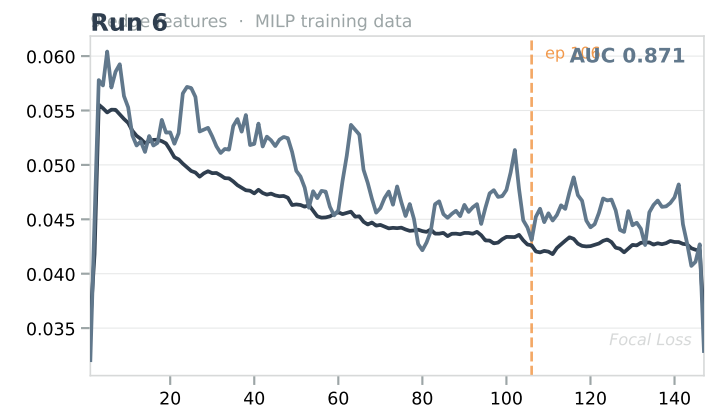
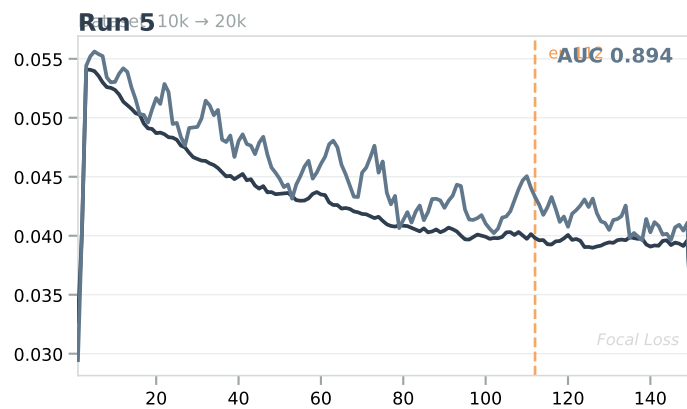
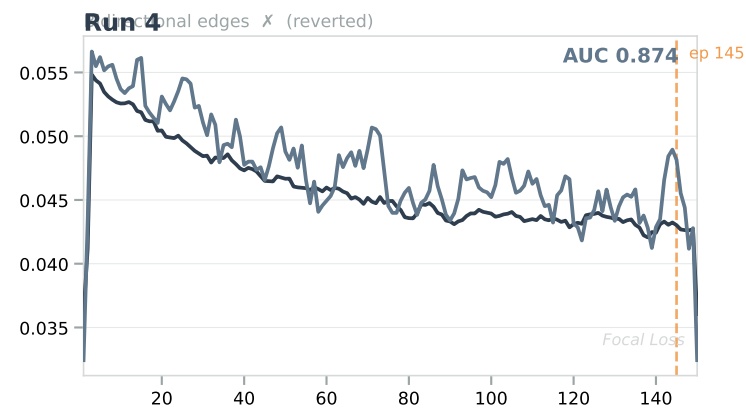
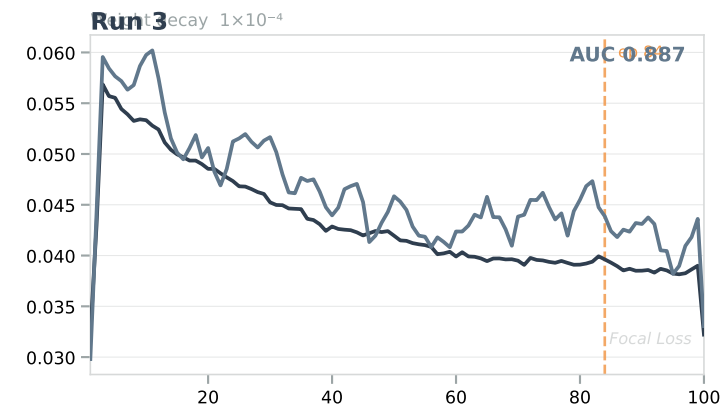
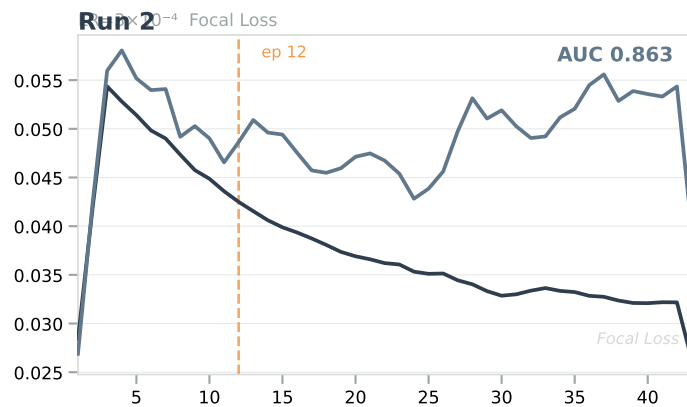
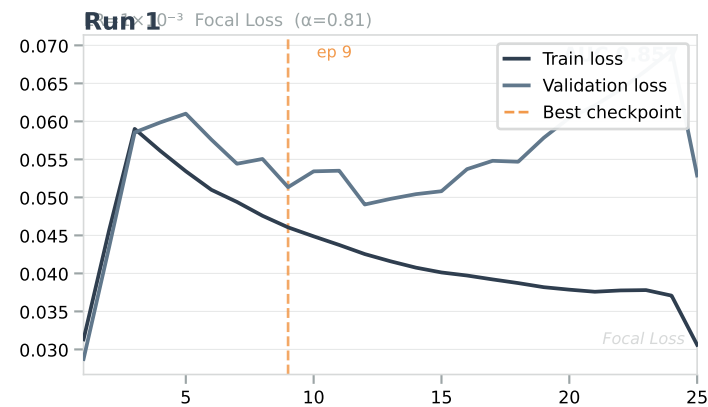


# GNN training progression — Runs 1-9



Runs 1-6: Focal Loss ( $\alpha=0.5-0.81$ ,  $\gamma=2.0$ ) · Runs 7-9: Weighted BCE (pos\_weight 4.1 → 2.5)